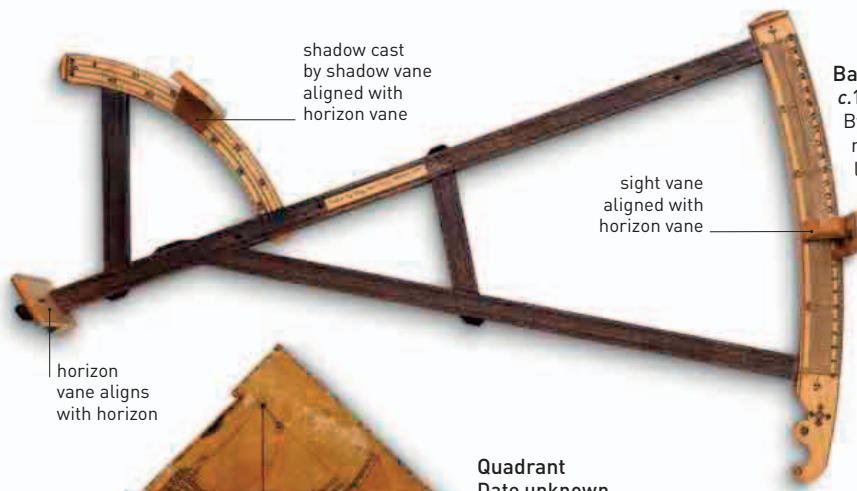




Backstaff
c.1700s

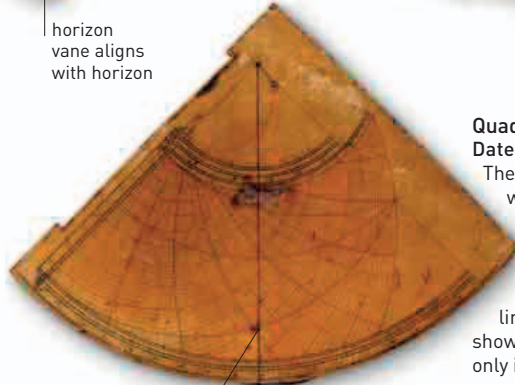
By the 18th century, navigators determined latitude by using a backstaff, which allowed them to determine the angle of the Sun without having to gaze directly at it.

sight vane aligned with horizon vane

horizon vane aligns with horizon

Quadrant
Date unknown

The quadrant was a simple way of determining latitude from the height of the Sun in the sky at noon. However, the plumb line that was needed to show vertical stayed steady only in still weather.

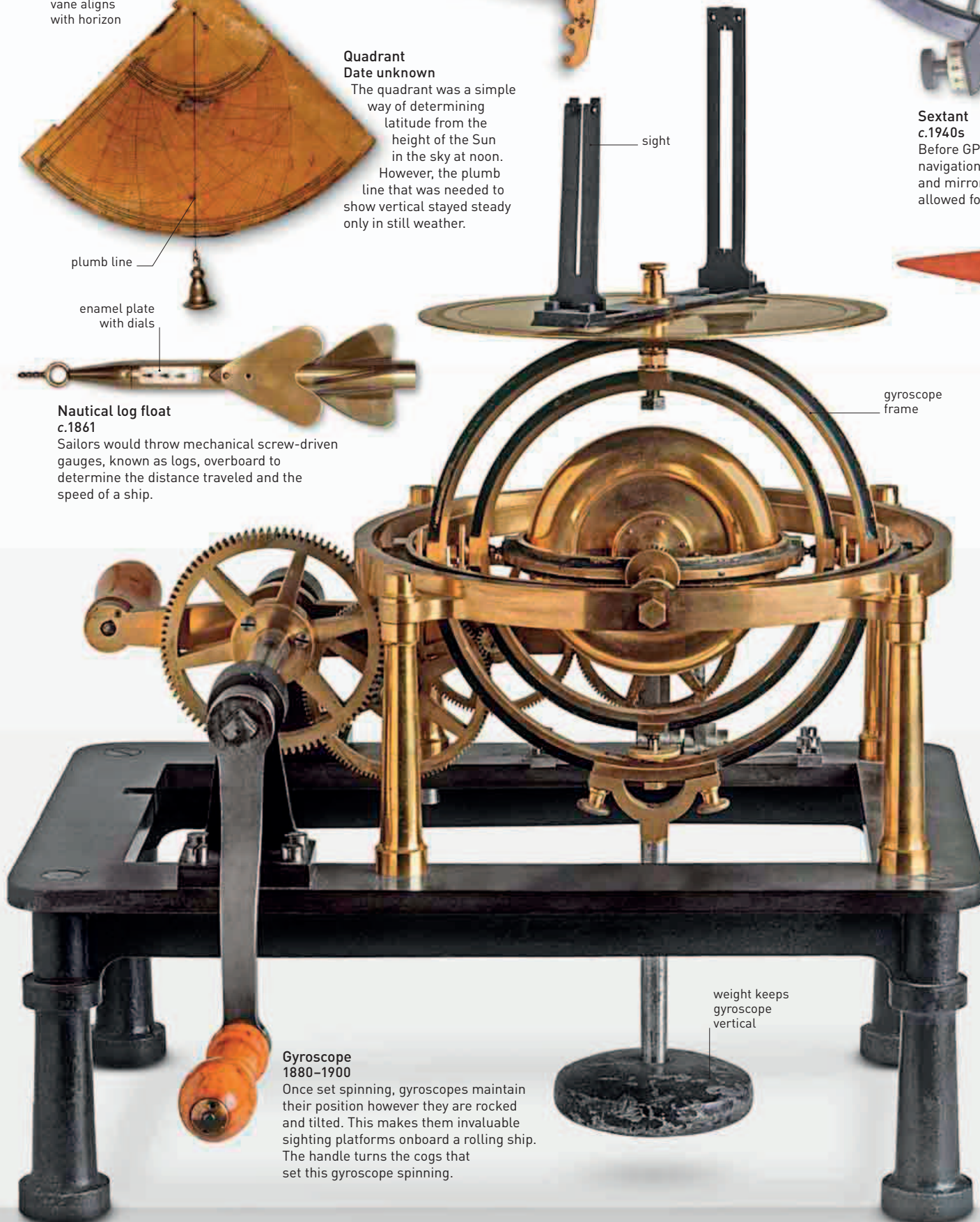


plumb line

enamel plate with dials

Nautical log float
c.1861

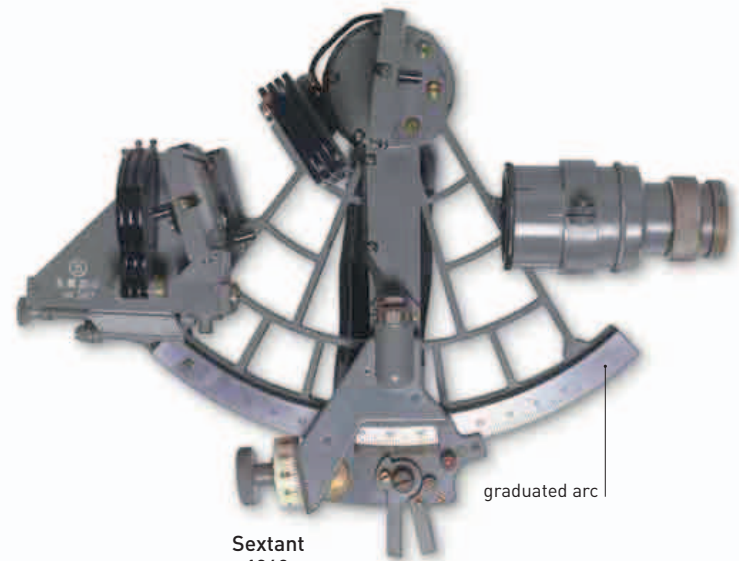
Sailors would throw mechanical screw-driven gauges, known as logs, overboard to determine the distance traveled and the speed of a ship.



Gyroscope
1880-1900

Once set spinning, gyroscopes maintain their position however they are rocked and tilted. This makes them invaluable sighting platforms onboard a rolling ship. The handle turns the cogs that set this gyroscope spinning.

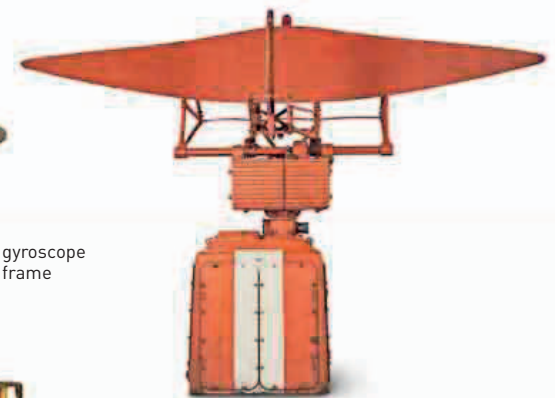
weight keeps gyroscope vertical



graduated arc

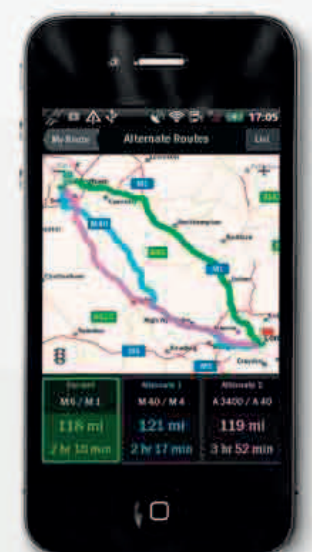
Sextant
c.1940s

Before GPS, the sextant was the ultimate navigation instrument. Its telescopic sights and mirrors for focusing the stars and Sun allowed for quick calculations of latitude.



Airfield radar dish
1953

Radars locate objects by bouncing radio waves off them, which aids navigation by giving aircraft accurate altitude readings.



GPS
c.2012

The global positioning system (GPS) of reference satellites provides an instant and accurate fix of position on even a hand-held device like this smartphone.